

STELLA VERKIJK

Research | Scientific Programming | Science Communication | Software Engineering

✉ s.verkijk@vu.nl

in stella-verkijk-3037621bb

🎓 Stella Verkijk

id 0009-0000-9263-2272

🔗 StellaVerkijk

👤 StellaVerkijk

PROFILE

I am a Language and Technology scientist specialised in the creation and evaluation of domain-specific Language Models. Deep Learning fascinates me, but within the field of Artificial Intelligence I have also investigated ontologies and Knowledge Representation. I have studied this broad range of Natural Language Processing techniques by implementing and evaluating them in highly specific domains, namely medical language data and archival material from the 17th century. I like building independent technology.

EDUCATION

PhD Candidate

2023-now

The Overlap and Divide Between Language-Derived Knowledge and World Knowledge

Vrije Universiteit Amsterdam: Computational Linguistics and Text Mining Lab

Advisor: Prof. Dr. Piek Vossen

Research Master

2019-2021

Human Language Technology (Cum Laude) Vrije Universiteit Amsterdam

Bachelors

Spanish Language and Culture (Cum Laude) Utrecht University

2017-2019

English Language and Culture Utrecht University

2015-2018

Exchange Program: Universidad de Chile

2018-2019

TECHNICAL SKILLS

Experienced with Python PyTorch Slurm Conda Tensorflow RDF Git **Familiar with** Lightning OWL

Languages: Dutch (native), English, Spanish (fluent), French (intermediate), German, Farsi (beginner)

PROJECTS

a-proof

a-proof creates structured information on human functioning from clinical text and conversation.

- helped set up and publish proof of concept
- oversaw annotation
- made MedRoBERTa.nl
- made sure we adhered to the hospital's privacy policy to get the model published open-source.

Globalise

Globalise creates a (re)search interface of the archives of the Dutch East India Company.

- designed event ontology & annotation scheme
- pre-trained models (GloBERTise)
- fine-tuned models for event classification and ran them on the 5 million scans in our corpus.

EXTRACURRICULAR

current

- **Public Engagement Fellow @ Amsterdam AI**

past

- **Advanced Scientific Programming in Python Summerschool**
- **Science Communication Lab**
- **Language Data Quality Lead Dutch @ Lausanne Business Solutions**

INTERESTS

Unraveling the Large Language Model hype | Dancing | Camping

PUBLICATIONS

- Verkijk, S.**, & Vossen, P. (Under submission). ROBE: Reversed-Order-Biased-Experts for Extracting Extreme Long-Tail Events.
- Goossens, H., **Verkijk, S.** & Vossen, P. (Under submission). Comparing Domain-Specific, Multilingual and Multitask Strategies to Overcome Data Scarcity for Semantic Role Labelling in Highly Noisy Early Modern Dutch
- Verkijk, S.**, & Vossen, P. (2026). Out-of-Tune rather than Fine-Tuned: How Pre-training, Fine-tuning and Tokenization Affect Semantic Similarity in a Historical, Non-Standardized Domain. In Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026) (pp. 515-531).
- Verkijk, S.**, Vossen, P., & Sommerauer, P. (2025). Language models lack temporal generalization and bigger is not better. In Findings of the Association for Computational Linguistics: ACL 2025 (pp. 20629-20637)
- Verkijk, S.**, & Vossen, P. (2025). Creating, anonymizing and evaluating the first medical language model pre-trained on Dutch Electronic Health Records: MedRoBERTa.nl. *Artificial Intelligence in Medicine*, 167, 1-13.
- Verkijk, S.**, Sommerauer, P., & Vossen, P. (2024). Studying Language Variation Considering the Re-Usability of Modern Theories, Tools and Resources for Annotating Explicit and Implicit Events in Centuries Old Text. In Proceedings of the Eleventh Workshop on NLP for Similar Languages, Varieties, and Dialects (VarDial 2024) (pp. 174-187).
- Verkijk, S.**, Roothaert, R., Pernisch, R., & Schlobach, S. (2023). Do you catch my drift? On the usage of embedding methods to measure concept shift in knowledge graphs. In Proceedings of the 12th Knowledge Capture Conference 2023 (pp. 70-74).
- Verkijk, S.**, & Vossen, P. (2023). Sunken Ships Shan't Sail: Ontology Design for Reconstructing Events in the Dutch East India Company Archives. In Proceedings of the Fourth Computational Humanities Conference (p. 320).
- Verkijk, S.**, & Vossen, P. (2022). Efficiently and Thoroughly Anonymizing a Transformer Language Model for Dutch Electronic Health Records: a Two-Step Method. In Proceedings of the Thirteenth Language Resources and Evaluation Conference (pp. 1098-1103).
- Meskers, c. G., van der Veen, S., Kim, J., Meskers, C. J., Smit, Q.T., **Verkijk, S.**, ... & van der Leeden, M. (2022). Automated recognition of functioning, activity and participation in COVID-19 from electronic patient records by natural language processing: a proof-of-concept. *Annals of Medicine*, 54(1), 235-243.
- Kim, J., **Verkijk, S.**, Gelein, E., van der Leeden, M., Meskers, C., Meskers, C., ... & Widdershoven, G. (2022). Modeling Dutch Medical Texts for Detecting Functional Categories and Levels of COVID-19 Patients. In Proceedings of the Thirteenth Language Resources and Evaluation Conference (pp. 4577-4585).
- Verkijk, S.**, & Vossen, P. (2021). MedRoBERTa.nl: A Language Model for Dutch Electronic Health Records. *Computational Linguistics in the Netherlands Journal*, 11, 141-159.

TALKS

- Verkijk, S.. (5 June 2026). Making Archives Accessible with Language Technology. Symposium Archivos del Mundo Hispánico. Utrecht, the Netherlands. Invited talk.
- Verkijk, S. & Vrooman, A. (30 April 2026). Workshop Outsmarting The Machine: Critically Evaluating Automatic Data Enrichments in Text. De Dag van de Nieuwste Geschiedenis, Antwerp, Belgium.
<https://doi.org/10.5281/zenodo.20118595>.
- Verkijk, S. (4 & 6 March 2026). Workshop Outsmarting the Machine: Critically Evaluating Automatic Data Enrichments in Text. GLOBALISE conference, Amsterdam, the Netherlands.
- Verkijk, S. (19 February 2026). Language Models Lack Temporal Generalization and Bigger is Not Better. DHLab Research Meeting, Amsterdam, the Netherlands. Invited talk.
- Verkijk, S. (27 November 2025). Applying LLMs to Archival Information Extraction. SSHOC-NL Workshop on Methodological reflections and practical applications of LLMs for the Social Sciences and Humanities, Amsterdam, the Netherlands. Invited talk.
- Verkijk, S. (24 November 2025). Language Models in the broader context of Artificial Intelligence . VSC member day, Kerkrade, the Netherlands. Invited talk.
- Verkijk, S. (17 October 2024). Automatic Event Detection in Early Modern Dutch (VOC) documents. Natural Language Processing (NLP) SIG, eScience Center, Amsterdam, the Netherlands. Invited talk.
- Verkijk, S & Smith, E. (12 November 2021) Workshop at Language & Technology; A Winning Combination. European Commission. Utrecht, the Netherlands. Invited.

TEACHING

at Vrije Universiteit Amsterdam

Text Mining Thesis Seminar. MA Course. Guest Lecturer.	2026
Advanced Machine Learning. MA Course. Guest Lecturer.	2023, 2024, 2025
Advanced Linguistics. MA Course. Co-teacher.	2023, 2024
Digital Humanities. MA Course. Guest Lecturer.	2023
Natural Language Processing Technologies. MSc Course. Teaching assistant.	2020
Machine Learning for Natural Language Processing. MA Course. Teaching assistant	2020